

Understanding Spans

Example:

$$V = \text{Poly}_{\leq 3}(\mathbb{R})$$

($\mathbb{R}$ -vector space)

$$X = \{x^2 + x, x^3 - x + 1\}$$

Q: $x^3 + x^2 + 1 \in \text{Span}(X)$? ?

A: Can we find $a_1, a_2 \in \mathbb{R}$
s.t.

$$a_1(x^2 + x) + a_2(x^3 - x + 1) \stackrel{?}{=} x^3 + x^2 + 1$$

$$a_2 x^3 + a_1 x^2 + a_1 x - a_2 x + a_2 \stackrel{?}{=} x^3 + x^2 + 1$$

$$a_2 x^3 + a_1 x^2 + (a_1 - a_2)x + \boxed{a_2}$$

Soln: $\boxed{a_2=1}, a_1=1$
 $a_1 - a_2 = 0, \boxed{a_2=1}$

Su: $x^3 + x^2 + 1 \in \text{Span}(X)$

But $x^3 + x^2 + 2 \notin \text{Span}(X)$

$\left[\begin{array}{l} a_2=1, a_1=1 \\ a_1 - a_2 = 0, a_2=2 \end{array} \right] \text{ INCONSISTENT}$

Rmk: When we arrived at these systems of equations, we "matched up" coeffs on both sides of an equality between vectors in $\text{Poly}_{\leq 3}(F)$.

$$\boxed{a_2 x^3 + a_1 x^2 + (a_1 - a_2)x + a_2} = \boxed{x^3 + x^2 + 1}$$

$$(c) \quad (a_2 - 1)x^3 + (a_1 - 1)x^2 + (a_1 - a_2)x + (a_2 - 1) \\ = 0$$

What we are actually using
is the following fact:

$$\text{If } c_3x^3 + c_2x^2 + c_1x + c_0 = 0$$

$$\text{then } c_3 = c_2 = c_1 = c_0 = 0$$

Linear independence

F: field V: F-v. space

Defⁿ A list of vectors

$$v_1, v_2, \dots, v_m \in V \text{ is}$$

linearly independent

$$\text{if } c_1v_1 + c_2v_2 + \dots + c_mv_m = 0, \text{ with } c_i \in F$$

$$\text{implies } c_1 = c_2 = \dots = c_m = 0$$

Otherwise, the list is linearly dependent

Example ①

$$1, x, x^2, x^3 \in \text{Poly}_{\leq 3}(F)$$

are linearly independent.

Rmk ① "List" instead of "set"
because we permit
repetitions of vectors

② A list $v_1, v_2, \dots, v_m \in V$
is linearly independent

$\iff$ (if & only if)

None of $v_1, v_2, \dots, v_m$ is in the
span of the remaining vectors

Pf:

$$\wedge C_1 v_1 + C_2 v_2 + \dots + C_m v_m = 0 \text{ but } C_1 \neq 0$$

$$\iff C_2 v_2 + \dots + C_m v_m = -C_1 v_1$$

$$\underbrace{(-c_1^{-1})}_{\text{scalar}} \cdot (-c_1^{-1}c_2v_2 + \dots + -c_1^{-1}c_mv_m) = V_1$$

$\Leftrightarrow V_1$ is a linear combination of $v_2, \dots, v_m$

$\Leftrightarrow V_1 \in \text{Span}(\{v_2, \dots, v_m\})$

We can think about this as follows: In a linearly independent list of vectors, each vector is giving us a new or independent direction in the vector space.

Prop (3) $v_1, v_2, \dots, v_m \in V$
and $v_k = 0$ for some k

Then the list is linearly dependent

Rmk (4) $v_1, v_2, \dots, v_m \in V$

$$\exists j \neq k \text{ s.t. } v_j = v_k$$

Then the list is linearly dependent

In other words, linearly independent cannot admit repeated vectors.

Example

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in F^3$$

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

This is $0 \Leftrightarrow c_1 = c_2 = c_3 = 0$

These vectors are therefore linearly independent.

Rmk (4) Suppose that

$$v_1, v_2, \dots, v_n \in V$$

are linearly independent

$$\& \quad w \in \text{Span}(\{v_1, v_2, \dots, v_n\})$$

Then

$$w = c_1 v_1 + c_2 v_2 + \dots + c_n v_n, \quad c_i \in F$$

where $c_1, c_2, \dots, c_n$ are

uniquely determined

Pf: If $w = d_1 v_1 + d_2 v_2 + \dots + d_n v_n$,
 $d_i \in F$

$$\Rightarrow C_1 v_1 + C_2 v_2 + \dots + C_m v_m$$

$$= d_1 v_1 + \dots + d_m v_m$$

$$\Rightarrow (C_1 - d_1) v_1 + \dots + (C_m - d_m) v_m$$

$$= 0$$

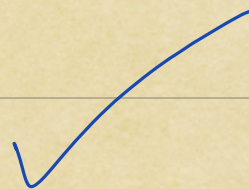
linear
independence

$$C_1 = d_1$$

$$C_2 = d_2$$

$\vdots$

$$C_m = d_m$$



Example:

Find $f(x) \in \text{Poly}_{\leq 3}(\mathbb{R})$

s.t. • $f(1) = 2$

• $f'(1) = \pi$

• $f^{(2)}(1) = e$

• $f^{(3)}(1) = \sqrt{2}$

We can simply write down
the Taylor expansion
at 1

$$\frac{\sqrt{2}}{3!} (x-1)^3 + \frac{e}{2!} (x-1)^2 + \pi(x-1) + 2$$